

Packet Tracer – QoS Basic Implementation (Packet Tracer 9.0 Compatible)

Objectives

Part 1: Build a QoS Test Topology

Part 2: Configure IP Addressing

Part 3: Verify Connectivity

Part 4: Classify Voice, Video, and Data Traffic

Part 5: Configure QoS Policies

Part 6: Apply QoS to an Interface

Part 7: Verify QoS Operation

Background / Scenario

You are the network administrator for a company that carries multiple traffic types across its network.

The network currently supports:

- Voice applications
- Video conferencing
- General data traffic

Management requires:

- Voice traffic receives the highest priority
- Video traffic receives medium priority
- Data traffic uses remaining bandwidth

Because Cisco Packet Tracer cannot generate realistic voice streams, a dedicated PC will simulate voice traffic.

This lab focuses on configuring and verifying QoS policies rather than measuring actual delay or jitter.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 1 Router (2911)
- 1 Switch (2960)
- 3 PCs
- 1 Server

Rename:

- Router0 → R1
- Switch0 → SW1
- PC0 → VOICE-PC
- PC1 → VIDEO-PC
- PC2 → DATA-PC
- Server0 → SERVER

Step 2: Connect Devices

Use Copper Straight-Through cables.

Device	Interface	Device	Interface
R1	G0/0	SW1	Fa0/24
VOICE-PC	Fa0	SW1	Fa0/1
VIDEO-PC	Fa0	SW1	Fa0/2
DATA-PC	Fa0	SW1	Fa0/3
SERVER	Fa0	SW1	Fa0/4

Wait until all links turn green.

Part 2: Configure IP Addressing

Network Plan

Device	IP Address	Mask	Gateway
VOICE-PC	192.168.10.20	255.255.255.0	192.168.10.1
VIDEO-PC	192.168.10.30	255.255.255.0	192.168.10.1
DATA-PC	192.168.10.40	255.255.255.0	192.168.10.1
SERVER	192.168.10.100	255.255.255.0	192.168.10.1
R1 G0/0	192.168.10.1	255.255.255.0	N/A

Configure Router

On R1:

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

end

Configure VOICE-PC

Desktop → IP Configuration

IP Address: 192.168.10.20

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Configure VIDEO-PC

IP Address: 192.168.10.30
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Configure DATA-PC

IP Address: 192.168.10.40
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Configure SERVER

IP Address: 192.168.10.100
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Part 3: Verify Connectivity

From each PC:

ping 192.168.10.100

Expected Result:

Reply from 192.168.10.100

All devices should communicate successfully.

Part 4: Create QoS Classification

Simulated Traffic Classes

Device	Traffic Type
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VOICE-PC	Voice
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VIDEO-PC	Video
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DATA-PC	Normal Data
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Create ACLs

On R1:

enable

configure terminal

access-list 101 permit ip host 192.168.10.20 any

access-list 102 permit ip host 192.168.10.30 any

end

Part 5: Create Class Maps

Voice Class

configure terminal

class-map match-any VOICE

match access-group 101

exit

Video Class

class-map match-any VIDEO

match access-group 102

exit

Part 6: Create QoS Policy

Create the policy map:

policy-map COMPANY-QOS

class VOICE

```
priority 100
```

```
class VIDEO  
bandwidth 50
```

```
class class-default  
fair-queue
```

```
exit
```

What These Commands Mean

Traffic Treatment

Voice Highest Priority

Video Guaranteed Bandwidth

Data Best Effort

Part 7: Apply QoS

Apply the policy to the router interface:

```
interface g0/0
```

```
service-policy output COMPANY-QOS
```

```
end
```

Verify QoS Configuration

Verify Class Maps

```
show class-map
```

Expected:

VOICE

VIDEO

Verify Policy Map

show policy-map

Expected:

Policy Map COMPANY-QOS

Verify Interface Policy

show policy-map interface g0/0

Expected:

Service-policy output: COMPANY-QOS

Verify Running Configuration

show running-config

Look for:

policy-map COMPANY-QOS

service-policy output COMPANY-QOS

Important Packet Tracer 9.0 Limitation

Even when QoS is configured correctly:

✓ Commands will apply successfully

✓ Class maps will exist

✓ Policy maps will exist

✓ Verification commands will work

But:

- ✗ Voice traffic will not actually sound better
- ✗ Video traffic will not visibly improve
- ✗ Packet Tracer does not simulate real congestion
- ✗ Jitter and latency are not accurately modeled

Therefore, the success criteria for this lab are:

- All devices can communicate
- QoS configuration is accepted
- Verification commands show the policy applied

Troubleshooting Tips

If QoS commands fail:

Verify router supports MQC:

`show version`

Verify ACLs:

`show access-lists`

Verify classes:

`show class-map`

Verify policy:

`show policy-map`

Verify application:

`show policy-map interface g0/0`

What You Learned

- What QoS is
- Why voice traffic requires prioritization
- How video traffic receives bandwidth guarantees

- How ACLs classify traffic
 - How Class Maps identify traffic categories
 - How Policy Maps prioritize traffic
 - How QoS is applied to interfaces
 - Packet Tracer QoS limitations
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Challenge Extensions (Optional)

Challenge 1: Add a Second Data PC

Objective

Simulate additional network traffic competing for bandwidth.

Instructions

Add:

Device	IP Address
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DATA-PC2	192.168.10.50
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Connect DATA-PC2 to SW1.

Configure:

IP Address: 192.168.10.50

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Verify:

ping 192.168.10.100

Expected Result:

Reply received

Challenge 2: Create a Business Application Class

Objective

Add a new QoS class for critical business applications.

Step 1: Create ACL

```
access-list 103 permit ip host 192.168.10.40 any
```

Step 2: Create Class Map

```
class-map match-any BUSINESS_APP  
match access-group 103
```

Step 3: Modify Policy Map

```
policy-map COMPANY-QOS
```

```
class VOICE  
priority 100
```

```
class VIDEO  
bandwidth 50
```

```
class BUSINESS_APP  
bandwidth 30
```

```
class class-default  
fair-queue
```

Verify:

```
show policy-map
```

Expected:

```
VOICE  
VIDEO  
BUSINESS_APP
```

Challenge 3: Increase Voice Priority

Objective

Give voice traffic even higher priority.

Modify:

```
policy-map COMPANY-QOS
```

```
class VOICE
```

```
priority 200
```

Verify:

```
show policy-map
```

Expected:

```
priority 200
```

Challenge 4: Add a Second Router

Objective

Create a routed network and apply QoS across a WAN link.

Add

Device IP

```
R2      10.10.10.2
```

Connect

```
R1 G0/1 ↔ R2 G0/0
```

Configure

R1:

```
interface g0/1
```

```
ip address 10.10.10.1 255.255.255.252
```

```
no shutdown
```

R2:

```
interface g0/0
```

```
ip address 10.10.10.2 255.255.255.252
```

```
no shutdown
```

Apply QoS on the inter-router link instead of the LAN interface.

Verify:

```
show policy-map interface g0/1
```

Challenge 5: Observe QoS Counters

Objective

Generate traffic and observe packet counters.

Generate multiple pings from:

- VOICE-PC
- VIDEO-PC
- DATA-PC

Run:

```
show policy-map interface g0/0
```

Expected:

Packets matched

Bytes matched

Counter values should increase.